

COMP5318/COMP4318 Machine Learning and Data Mining

Semester 1, 2025

Information about the exam

- The exam is in-person, paper-based
- Permitted materials:
 - One sheet of your own notes - double-sided A4 size, typed or handwritten
 - Calculator - handheld, non-programmable
- No other materials or devices are allowed
- You will need to leave the A4-sheet with the exam paper, so it is recommended that you make a copy of this sheet for your own records
- You need to have a photo identification - Student ID card or other accepted documents
- The exam paper is like a booklet. You write your answers on the exam paper, in the space provided. There is a question, then a space for you to write the answer; another question and space for the answer, etc. To write the answers you should use black or blue pen, not pencil.
- The exam paper is **confidential**. You must not discuss the exam questions with other people, post or distribute the exam questions in any way **after the exam**.
- The duration of the exam is standard: 2 hours + 10 minutes reading time.
- The exam is worth 100 marks (=60% of your final mark). To pass the course you need at least 40% on the exam (i.e. 40 marks), regardless of what your mark during the semester is. This is a school rule.
- There are 3 types of questions: 1) multiple-choice questions (about 20% of the exam), 2) questions requiring short answers, 3) problem-solving/calculation questions.

Exam cover page



THE UNIVERSITY OF
SYDNEY

Room Number _____

Seat Number _____

Student Number

ANONYMOUSLY MARKED

(Please do not write your name on this exam paper)

CONFIDENTIAL EXAM PAPER

This paper is not to be removed from the exam venue

Computer Science

EXAMINATION

Semester 1 – Final, 2025

COMP4318/COMP5318 Machine Learning and Data Mining

For Examiner Use Only

EXAM WRITING TIME: 2 hours

READING TIME: 10 minutes

EXAM CONDITIONS:

This is a RESTRICTED OPEN book exam - specified materials permitted

MATERIALS PERMITTED IN THE EXAM VENUE:

(No electronic aids are permitted e.g. laptops, phones)

Calculator - non-programmable

One A4 sheet of handwritten and/or typed notes double-sided

MATERIALS TO BE SUPPLIED TO STUDENTS:

None

INSTRUCTIONS TO STUDENTS:

Answer all questions using blue or black pen (not pencil), on this examination paper in the spaces provided.

Q	Mark
1 /26	
2 /10	
3 /10	
4 /12	
5 /14	
6 /8	
7 /4	
8 /6	
9 /10	

Total _____/100

Please tick the box to confirm that your examination paper is complete. ☐

Sample exam questions

Question 1. (Multiple choice question)

Select the correct answer.

1. Leave-one-out cross validation is suitable for large data sets.

- a) True
- b) False

2. The regression line minimizes the sum of the residuals

- a) True
- b) False

3. A single perceptron can solve the XOR problem.

True False

Question 2. (Short answer question)

1. Why do we need to apply normalization when using distance-based algorithms such as k-Nearest Neighbor?

2. In linear support vector machines, we use dot products both during training and during classification of a new example. What vectors are these products of?

During training:

During classification of new example:

3. List one disadvantage of applying a multi-layer perceptron neural network to perform handwritten digits image classification.

4. When we use k-Nearest Neighbor for regression tasks (the class value is numeric), how is the prediction for new examples calculated?

Calculation (problem solving) questions

Question 3. Decision tree

Given is the following training data where **location**, **weather** and **expensive** are the features and **holiday** is the class.

location	weather	expensive	holiday
nice	sunny	Y	good
nice	sunny	N	bad
boring	rainy	Y	good
boring	sunny	N	bad
nice	rainy	Y	good
boring	rainy	N	good
boring	rainy	N	good

- What is the entropy of this set of training examples with respect to the class?
- We would like to build a decision tree using information gain. Which attribute will be selected as a root of the tree? Show your calculations.

You may use this table:

x	y	$-(x/y) * \log_2(x/y)$	x	y	$-(x/y) * \log_2(x/y)$
1	2	0.50	1	6	0.43
1	3	0.53	5	6	0.22
2	3	0.39	1	7	0.40
1	4	0.5	2	7	0.52
3	4	0.31	3	7	0.52
1	5	0.46	4	7	0.46
2	5	0.53	5	7	0.35
3	5	0.44	6	7	0.19
4	5	0.26			

Question 4. Naïve Bayes

Given is the following training data where **location**, **weather**, **companion** and **expensive** are the features and **holiday** is the class.

location	weather	companion	expensive	holiday
nice	sunny	annoying	Y	good
nice	sunny	annoying	N	bad
boring	rainy	great	Y	good
boring	sunny	great	Y	bad
nice	rainy	great	Y	good
boring	rainy	annoying	N	good
boring	rainy	great	N	good

Use Naïve Bayes to predict the value of **holiday** for the following new example, showing your calculations:
location=boring, weather=sunny, companion=annoying, expensive=Y.

Question 5. 1R

Given the training data in the table below where **credit history**, **debt**, **deposit** and **income** are attributes and **risk** is the class, predict the class of the following new example using the 1R algorithm: *credit history=unknown, debt=low, deposit=none, income=average*. If needed, settle ties by random selection. Show your calculations.

credit history	debt	deposit	income	risk
bad	high	none	low	high
unknown	high	none	average	high
unknown	low	none	average	moderate
unknown	low	none	low	high
unknown	low	none	high	low
unknown	low	adequate	high	low
bad	low	none	low	high
bad	low	adequate	high	moderate
good	low	none	high	low
good	high	adequate	high	low
good	high	none	low	high
good	high	none	average	moderate
good	high	none	high	low
bad	high	none	average	high

Question 6. Perceptron

Given is the following training set:

	input	output
ex. 1:	1 0 0	1
ex. 2:	0 1 1	0
ex. 3:	1 1 0	1
ex. 4:	1 1 1	0
ex. 5:	0 0 1	0

a) Train a perceptron **with a bias** on this training set. Assume that all initial weights (including the bias of the neuron) are 0. Show the set of weights (including the bias) at the end of the first epoch. Apply the examples in the given order.

Recall that the perceptron uses a step function defined as:

$\text{step}(n) = 1$, if $n \geq 0$
= 0, otherwise.

Question 7. K-means clustering

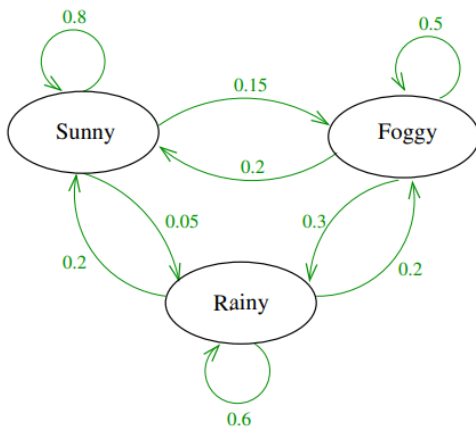
Suppose that we are given 7 examples to cluster: A, B, C, D, E, F and G. The distance between them is given by the following matrix:

	A	B	C	D	E	F	G
A	0	10	2	1	12	5	4
B	10	0	4	3	6	23	7
C	2	4	0	5	9	14	19
D	1	3	5	0	1	7	4
E	12	6	9	1	0	2	18
F	5	23	14	7	2	0	6
G	4	7	19	4	18	6	0

Run the **k-means** algorithm to group these examples into 2 clusters for 1 epoch. The initial centroids are A and B. Show the resulting clusters.

Question 8. Markov models

Given is the following Markov model for the weather in Sydney:



a) Given that today the weather is *Sunny*, what is the probability that it will be *Sunny* tomorrow and *Rainy* the day after tomorrow, i.e. what is the probability $P(\pi_3 = \text{Rainy}, \pi_2 = \text{Sunny} | \pi_1 = \text{Sunny})$?

Hint: $P(A,B|C) = P(A|B,C) P(B|C)$

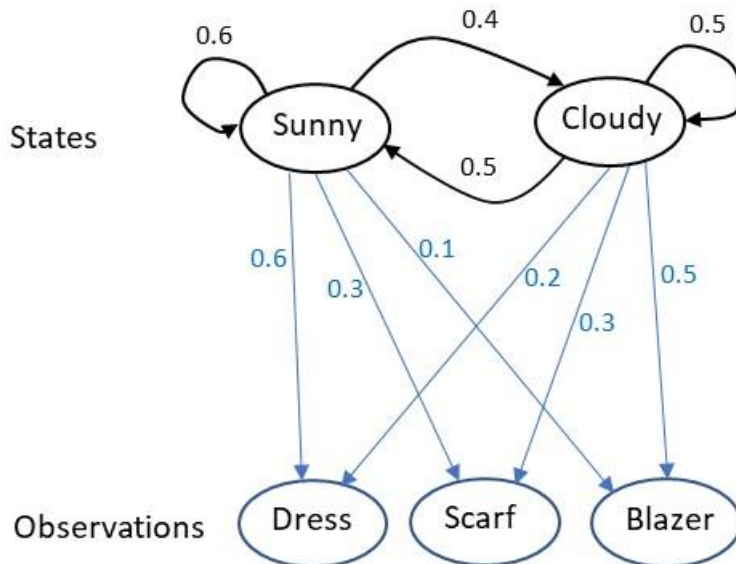
b) If the weather yesterday was *Rainy*, and today is *Foggy*, what is the probability that tomorrow it will be *Sunny*?

For both questions, briefly show your calculations.

Question 9. Hidden Markov models

Julia tested positive to COVID and had to quarantine at home for several days. Her friend Nicole came to bring her food every day. We don't know what the weather was on the quarantine days but we know the type of clothing Nicole wore and it provides evidence about the weather.

The following Hidden Markov Model models the situation. The initial state probabilities are: $A_0(\text{Sunny})=0.5$ and $A_0(\text{Cloudy})=0.5$.



Suppose that on the first quarantine day Nicole wore a dress and on the second she wore a blazer.

- What is the probability of the observation sequence?
- What is the most likely sequence of hidden states?

Briefly show your calculations.